

# Internal, near-surface, and scattered electromagnetic fields for a layered spheroid with arbitrary illumination

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A spheroidal coordinate separation-of-variables solution has been developed for the determination of internal, near-surface, and scattered electromagnetic fields of a layered spheroid (either prolate or oblate) with arbitrary monochromatic illumination (e.g., plane wave or focused Gaussian beam). Calculated results are presented for layered 2:1 axis ratio prolate and oblate spheroids with an equivalent sphere size parameter of 20. © 2001 Optical Society of America

OCIS codes: 290.4020, 350.3950, 280.1100, 260.2110.

## 1. Introduction

In an earlier paper,<sup>1</sup> a spheroidal coordinate separation-of-variables solution was developed for the determination of internal and external electromagnetic fields for a homogeneous spheroid, of either prolate or oblate geometry, illuminated by monochromatic light of arbitrary character (plane wave, focused beam, etc.). In this paper, the procedure developed in Ref. 1 is generalized so as to obtain a corresponding solution for a spheroid consisting of a homogeneous core surrounded by a single homogeneous layer. This development could be used, for example, to determine the electromagnetic fields that result from the interaction of a focused laser beam with an appreciably elongated (prolatelike) or appreciably flattened (oblatelike) solid particle coated with a layer of condensed liquid or with a spheroid-shaped biological particle that has been approximately modeled as a homogeneous nucleus surrounded by a homogeneous layer of cytoplasm.

The problem of plane-wave scattering by a layered spheroid has been considered previously. Wang and Barber<sup>2</sup> solved the problem by applying the extended boundary condition method in spherical coordinates, whereas Sebak and Sinha,<sup>3</sup> Som-sikov,<sup>4</sup> and Farafonov *et al.*<sup>5-7</sup> have solved the

problem using a spheroidal coordinate separation-of-variables method. The research presented here differs from these previous efforts, however, in that the incident illumination is not restricted to the plane wave, but can be of any arbitrary form including, for example, a focused Gaussian beam or scattered light from an adjacent particle or surface.<sup>8-10</sup> In addition, in Refs. 2-7 only far-field scattering results are presented. In this paper, internal and near-surface electromagnetic field distributions are also presented. A study of the internal and near-surface electromagnetic field distributions can assist in the development of a physical interpretation of the origins of the resultant far-field scattering. For high-intensity laser applications, the internal and near-surface electromagnetic field distributions can be used to determine the location within the particle where maximum heating occurs or the location where electrical breakdown would initiate.

Internal and near-surface electromagnetic fields for a layered spheroid with arbitrary illumination could also be determined with the spherical coordinate boundary-matching method described in Ref. 11. As discussed in Ref. 11, however, this method is limited to near-spherical geometries and could be used only for spheroids with axis ratios up to approximately 1.40. The spheroidal coordinate method presented here is applicable for high (as well as low) axis ratio spheroids.

## 2. Theoretical Development

### A. General Theory

The electromagnetic interaction of a known incident monochromatic field with a layered spheroid consisting of a core surrounded by a single layer is consid-

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Received 12 September 2000; revised manuscript received 23 April 2001.

0003-6935/01/213598-10\$15.00/0

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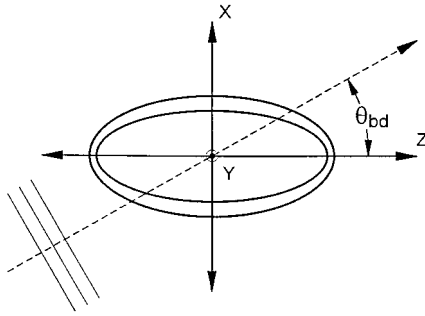


Fig. 1. Schematic of the geometric arrangement. The boundary of the spheroid rotates about the  $z$  axis. For the prolate spheroid (as shown), the major axis of the spheroid is along the  $z$  axis. For the oblate spheroid, the major axis of the spheroid is along the  $x$  axis. The outer surface of the spheroid (the layer-external interface) is located at  $\xi = \xi_0$  and the core-layer interface is located at  $\xi = \xi_c < \xi_0$ .

ered. A schematic of the assumed geometry is shown in Fig. 1. The spheroid core, spheroid layer, and the surrounding medium are each homogeneous, isotropic, and nonmagnetic ( $\mu = 1$ ); and the surrounding medium is assumed to be infinite and non-absorbing. An implicit time dependence of  $\exp(-i\omega t)$  is assumed, and the electromagnetic field is nondimensionalized relative to an electric field amplitude  $E_0$  associated with the incident field.

A spheroid is created when an ellipse is rotated about its axis. If the ellipse is rotated about its major axis, a prolate spheroid is formed. If the ellipse is rotated about its minor axis, an oblate spheroid is formed. The spheroid has an associated semifocal length  $f$ . If  $a$  is the length of the semimajor axis and  $b$  is the length of the semiminor axis of the spheroid, then

$$f = a[1 - (b/a)^2]^{1/2}. \quad (1)$$

In the following, the rectangular coordinates  $(x, y, z)$ , the spatial position vector  $\mathbf{r}$ , and the vector calculus operators  $(\nabla, \nabla^2)$  are all nondimensionalized relative to the semifocal length  $f$ .

The eigenfunctions of the separation-of-variables solution of the scalar Helmholtz equation,

$$(\nabla^2 + h^2)\Pi = 0, \quad (2)$$

in spheroidal coordinates  $(\eta, \xi, \phi)$  are of the form

$$\Pi_{lm} = S_{lm}(h, \eta) R_{lm}(h, \xi) \exp(im\phi), \quad (3)$$

where  $h = kf$ ,  $k$  is the wave number,  $S_{lm}$  is the spheroidal angle function,  $R_{lm}$  is the spheroidal radial function (which may be of the first kind,  $R_{lm}^{(1)}$ ; second kind,  $R_{lm}^{(2)}$ ; third kind,  $R_{lm}^{(3)} = R_{lm}^{(1)} + iR_{lm}^{(2)}$ ; or fourth kind,  $R_{lm}^{(4)} = R_{lm}^{(1)} - iR_{lm}^{(2)}$ ), and the azimuthal angle dependence is contained in the complex exponential  $\exp(im\phi)$ .  $l$  and  $m$  are integer indices with  $-\infty < m < \infty$  and, for a given  $m$ ,  $|m| \leq l < \infty$ .  $\eta$  is the spheroidal angle coordinate,  $\xi$  is the spheroidal radial coordinate, and  $\phi$  is the azimuthal coordinate. The  $S_{lm}$  and  $R_{lm}$  functions are different depending on

whether a prolate or oblate spheroidal coordinate system is chosen. A computer program for generating the associated eigenvalues,  $\lambda_{lm}(h)$ , and subroutines for generating the  $S_{lm}(h, \eta)$  and  $R_{lm}(h, \xi)$  functions were written based on procedures described in Flammer.<sup>1,12,13</sup>

Within a source-free, homogeneous medium, both the electric field  $\mathbf{E}$  and the magnetic field  $\mathbf{H}$  are solenoidal ( $\nabla \cdot \mathbf{E} = 0$ ,  $\nabla \cdot \mathbf{H} = 0$ ) and satisfy the vector Helmholtz equations

$$(\nabla^2 + h^2)\mathbf{E} = 0, \quad (4)$$

$$(\nabla^2 + h^2)\mathbf{H} = 0. \quad (5)$$

Following the approach of Stratton,<sup>14</sup> independent solenoidal vector eigenfunction solutions  $(\mathbf{M}_{lm}, \mathbf{N}_{lm})$  of the vector Helmholtz equation can be obtained from the corresponding scalar eigenfunctions by the following vector operations:

$$\mathbf{M}_{lm} = \nabla \times (\mathbf{r}\Pi_{lm}), \quad (6)$$

$$\mathbf{N}_{lm} = \frac{1}{h} \nabla \times \mathbf{M}_{lm}. \quad (7)$$

In spheroidal coordinates, a spheroid is formed by a surface of constant radial coordinate ( $\xi = \text{constant}$ ); thus a specified value of  $\xi_0$  is used to define the outer boundary of the spheroid, and a specified value of  $\xi_c < \xi_0$  is used to define the boundary of the core-layer interface. The external particle ( $\xi > \xi_0$ ) electromagnetic field consists of a sum of the known incident field indicated by the superscript ( $i$ ) and the to-be-determined scattered field indicated by the superscript ( $s$ ). The electromagnetic field within the particle core ( $\xi < \xi_c$ ) indicated by the superscript ( $c$ ) and the electromagnetic field within the particle layer ( $\xi_c < \xi < \xi_0$ ) indicated by the superscript ( $l$ ) are also to be determined. Expressions for the scattered and internal fields are developed when we take summations over the appropriate forms of the corresponding vector eigenfunctions as follows.

For the scattered field,

$$\mathbf{E}^{(s)} = \sum_{l,m} [a_{lm}\mathbf{N}_{lm}^{(s)} + b_{lm}\mathbf{M}_{lm}^{(s)}], \quad (8)$$

$$\mathbf{H}^{(s)} = -i\sqrt{\epsilon_{\text{ext}}} \sum_{l,m} [a_{lm}\mathbf{M}_{lm}^{(s)} + b_{lm}\mathbf{N}_{lm}^{(s)}], \quad (9)$$

where

$$\Pi_{lm}^{(s)} = S_{lm}(h_{\text{ext}}, \eta) R_{lm}^{(3)}(h_{\text{ext}}, \xi) \exp(im\phi). \quad (10)$$

For the core field,

$$\mathbf{E}^{(c)} = \sum_{l,m} [c_{lm}\mathbf{N}_{lm}^{(c)} + d_{lm}\mathbf{M}_{lm}^{(c)}], \quad (11)$$

$$\mathbf{H}^{(c)} = -i\sqrt{\epsilon_{\text{ext}}} n_c \sum_{l,m} [c_{lm}\mathbf{M}_{lm}^{(c)} + d_{lm}\mathbf{N}_{lm}^{(c)}], \quad (12)$$

where

$$\Pi_{lm}^{(c)} = S_{lm}(h_c, \eta) R_{lm}^{(1)}(h_c, \xi) \exp(im\phi). \quad (13)$$

For the layer field,

$$\mathbf{E}^{(l)} = \sum_{l,m} [e_{lm} \mathbf{N}_{lm}^{(l,1)} + f_{lm} \mathbf{M}_{lm}^{(l,1)} + g_{lm} \mathbf{N}_{lm}^{(l,2)} + h_{lm} \mathbf{M}_{lm}^{(l,2)}], \quad (14)$$

$$\mathbf{H}^{(l)} = -i \sqrt{\epsilon_{\text{ext}}} n_l \sum_{l,m} [e_{lm} \mathbf{M}_{lm}^{(l,1)} + f_{lm} \mathbf{N}_{lm}^{(l,1)} + g_{lm} \mathbf{M}_{lm}^{(l,2)} + h_{lm} \mathbf{N}_{lm}^{(l,2)}], \quad (15)$$

where

$$\Pi_{lm}^{(l,1)} = S_{lm}(h_l, \eta) R_{lm}^{(1)}(h_l, \xi) \exp(im\phi), \quad (16)$$

$$\Pi_{lm}^{(l,2)} = S_{lm}(h_l, \eta) R_{lm}^{(2)}(h_l, \xi) \exp(im\phi). \quad (17)$$

In Eqs. (8)–(17),  $\epsilon_{\text{ext}}$  is the dielectric constant of the surrounding (external) medium,  $n_c = \sqrt{\epsilon_c/\epsilon_{\text{ext}}}$  is the relative index of refraction of the core, and  $n_l = \sqrt{\epsilon_l/\epsilon_{\text{ext}}}$  is the relative index of refraction of the layer. Thus  $h_{\text{ext}} = k_{\text{ext}} f = (2\pi/\lambda_{\text{ext}}) f$ ,  $h_c = k_c f = n_c h_{\text{ext}}$ , and  $h_l = k_l f = n_l h_{\text{ext}}$ , where  $\lambda_{\text{ext}}$  is the wavelength in the surrounding medium. For the scattered field, only the  $R_{lm}^{(3)}$  spheroidal radial function is included in the expression for  $\Pi_{lm}^{(s)}$  because, in the limit of large  $\xi$ , this function corresponds to an outgoing traveling wave, appropriate for the scattered field solution. [In the same limit, the  $R_{lm}^{(4)}$  spheroidal wave function would correspond to an incoming traveling wave.] For the core field, only the  $R_{lm}^{(1)}$  spheroidal radial function is included in the expression for  $\Pi_{lm}^{(c)}$  because  $R_{lm}^{(2)}$  is unbounded as  $\xi \rightarrow 0$ . In practice, the double summation indices are truncated such that  $-M < m < +M$  and  $|m| \leq l < L$ , where  $M$  and  $L$  are chosen sufficiently large for convergence of the solution.

The electromagnetic field components of the incident field  $\mathbf{E}^{(i)}$  and  $\mathbf{H}^{(i)}$  (plane wave, focused beam, etc.) are assumed known. The unknown expansion coefficients ( $a_{lm}$ ,  $b_{lm}$ ,  $c_{lm}$ ,  $d_{lm}$ ,  $e_{lm}$ ,  $f_{lm}$ ,  $g_{lm}$ ,  $h_{lm}$ ) are determined when we apply the boundary conditions of continuity of the tangential electromagnetic field over the outer surface of the particle ( $\xi = \xi_0$ ) and over the core–layer interface ( $\xi = \xi_c$ ). At  $\xi = \xi_0$ ,

$$E_{\eta}^{(l)} - E_{\eta}^{(s)} = E_{\eta}^{(i)}, \quad (18)$$

$$E_{\phi}^{(l)} - E_{\phi}^{(s)} = E_{\phi}^{(i)}, \quad (19)$$

$$H_{\eta}^{(l)} - H_{\eta}^{(s)} = H_{\eta}^{(i)}, \quad (20)$$

$$H_{\phi}^{(l)} - H_{\phi}^{(s)} = H_{\phi}^{(i)}; \quad (21)$$

and at  $\xi = \xi_c$ ,

$$E_{\eta}^{(l)} - E_{\eta}^{(c)} = 0, \quad (22)$$

$$E_{\phi}^{(l)} - E_{\phi}^{(c)} = 0, \quad (23)$$

$$H_{\eta}^{(l)} - H_{\eta}^{(c)} = 0, \quad (24)$$

$$H_{\phi}^{(l)} - H_{\phi}^{(c)} = 0. \quad (25)$$

Next, the series expansions of Eqs. (8)–(17) are used to substitute for the scattered, layer, and core electromagnetic field components in the eight boundary

condition equations. The resultant series-expansion forms of Eqs. (18)–(21) are then each multiplied by  $S_{l'm'}(h_{\text{ext}}, \eta) \exp(-im'\phi)$  and integrated over the outer surface of the spheroid ( $\xi = \xi_0$ ). Similarly, the resultant series-expansion forms of Eqs. (22)–(25) are each multiplied by  $S_{l'm'}(h_l, \eta) \exp(-im'\phi)$  and integrated over the surface of the core–layer interface ( $\xi = \xi_c$ ).

For a maximum summation limit  $L$  and for a particular value of  $m$ , we obtain the following set of simultaneous linear algebraic equations:

$$\sum_{l=|m|}^L (-I_{lm'l}^1 a_{lm} - I_{lm'l}^2 b_{lm} + I_{lm'l}^3 e_{lm} + I_{lm'l}^4 f_{lm} + I_{lm'l}^5 g_{lm} + I_{lm'l}^6 h_{lm}) = \frac{1}{2\pi} A_{l'm}^{\eta}, \quad (26)$$

$$\sum_{l=|m|}^L (-I_{lm'l}^7 a_{lm} - I_{lm'l}^8 b_{lm} + I_{lm'l}^9 e_{lm} + I_{lm'l}^{10} f_{lm} + I_{lm'l}^{11} g_{lm} + I_{lm'l}^{12} h_{lm}) = \frac{1}{2\pi} A_{l'm}^{\phi}, \quad (27)$$

$$\sum_{l=|m|}^L (-I_{lm'l}^2 a_{lm} - I_{lm'l}^1 b_{lm} + n_l I_{lm'l}^4 e_{lm} + n_l I_{lm'l}^3 f_{lm} + n_l I_{lm'l}^6 g_{lm} + n_l I_{lm'l}^5 h_{lm}) = \frac{i}{2\pi \sqrt{\epsilon_{\text{ext}}}} B_{l'm}^{\eta}, \quad (28)$$

$$\sum_{l=|m|}^L (-I_{lm'l}^8 a_{lm} - I_{lm'l}^7 b_{lm} + n_l I_{lm'l}^{10} e_{lm} + n_l I_{lm'l}^9 f_{lm} + n_l I_{lm'l}^{12} g_{lm} + n_l I_{lm'l}^{11} h_{lm}) = \frac{i}{2\pi \sqrt{\epsilon_{\text{ext}}}} B_{l'm}^{\phi}, \quad (29)$$

$$\sum_{l=|m|}^L (-I_{lm'l}^{13} c_{lm} - I_{lm'l}^{14} d_{lm} + I_{lm'l}^{15} e_{lm} + I_{lm'l}^{16} f_{lm} + I_{lm'l}^{17} g_{lm} + I_{lm'l}^{18} h_{lm}) = 0, \quad (30)$$

$$\sum_{l=|m|}^L (-I_{lm'l}^{19} c_{lm} - I_{lm'l}^{20} d_{lm} + I_{lm'l}^{21} e_{lm} + I_{lm'l}^{22} f_{lm} + I_{lm'l}^{23} g_{lm} + I_{lm'l}^{24} h_{lm}) = 0, \quad (31)$$

$$\sum_{l=|m|}^L (-n_c I_{lm'l}^{14} c_{lm} - n_c I_{lm'l}^{13} d_{lm} + n_l I_{lm'l}^{16} e_{lm} + n_l I_{lm'l}^{15} f_{lm} + n_l I_{lm'l}^{18} g_{lm} + n_l I_{lm'l}^{17} h_{lm}) = 0, \quad (32)$$

$$\sum_{l=|m|}^L (-n_c I_{lm'l}^{20} c_{lm} - n_c I_{lm'l}^{19} d_{lm} + n_l I_{lm'l}^{22} e_{lm} + n_l I_{lm'l}^{21} f_{lm} + n_l I_{lm'l}^{24} g_{lm} + n_l I_{lm'l}^{23} h_{lm}) = 0, \quad (33)$$

for all  $l' = |m|$  to  $L$ . The one-dimensional integrals  $I_{lm'l'}^{1-24}$  are dependent solely on the geometry of the particle ( $\xi_0$ ,  $\xi_c$ ) and are defined in Appendix A. The surface integrals  $A_{l'm}^{\eta}$ ,  $A_{l'm}^{\phi}$ ,  $B_{l'm}^{\eta}$ , and  $B_{l'm}^{\phi}$  are depen-

dent on both the geometry of the particle ( $\xi_0$ ) and the incident field [ $\mathbf{E}^{(i)}$ ,  $\mathbf{H}^{(i)}$ ] and are defined as follows:

$$A_{l'm}^\eta = \int_0^{2\pi} \int_{-1}^1 E_\eta^{(i)}(\xi_0, \eta, \phi) S_{l'm}(h_{\text{ext}}, \eta) \times \exp(-im\phi) d\eta d\phi, \quad (34)$$

$$A_{l'm}^\phi = \int_0^{2\pi} \int_{-1}^1 E_\phi^{(i)}(\xi_0, \eta, \phi) S_{l'm}(h_{\text{ext}}, \eta) \times \exp(-im\phi) d\eta d\phi, \quad (35)$$

$$B_{l'm}^\eta = \int_0^{2\pi} \int_{-1}^1 H_\eta^{(i)}(\xi_0, \eta, \phi) S_{l'm}(h_{\text{ext}}, \eta) \times \exp(-im\phi) d\eta d\phi, \quad (36)$$

$$B_{l'm}^\phi = \int_0^{2\pi} \int_{-1}^1 H_\phi^{(i)}(\xi_0, \eta, \phi) S_{l'm}(h_{\text{ext}}, \eta) \times \exp(-im\phi) d\eta d\phi, \quad (37)$$

For a particular value of  $m$ , Eqs. (26)–(33) thus provide 8 ( $L - |m| + 1$ ) linear algebraic equations to solve for the 8 ( $L - |m| + 1$ ) values of ( $a_{lm}$ ,  $b_{lm}$ ,  $c_{lm}$ ,  $d_{lm}$ ,  $e_{lm}$ ,  $f_{lm}$ ,  $g_{lm}$ ,  $h_{lm}$ ). Note that, for the special case of  $m = 0$ , the one-dimensional integrals  $I^2$ ,  $I^4$ ,  $I^6$ ,  $I^7$ ,  $I^9$ ,  $I^{11}$ ,  $I^{14}$ , and  $I^{16}$  are zero, so Eqs. (26), (29), (30), and (33) can be used to solve for the 4 ( $L + 1$ ) values of ( $a_{l0}$ ,  $c_{l0}$ ,  $e_{l0}$ ,  $g_{l0}$ ); and Eqs. (27), (28), (31), and (32) can be used to solve for the 4 ( $L + 1$ ) values of ( $b_{l0}$ ,  $d_{l0}$ ,  $f_{l0}$ ,  $h_{l0}$ ).

In summary, the procedure for determining the internal and external electromagnetic fields for a known field incident on a layered spheroid is as follows. For a given set of input parameters ( $\xi_0$ ,  $\xi_c$ ,  $h_{\text{ext}}$ ,  $n_l$ ,  $n_c$ , incident field parameters, etc.), the  $I_{lm'l'}$  one-dimensional integrals of Appendix A and the  $A_{l'm}^\eta$ ,  $A_{l'm}^\phi$ ,  $B_{l'm}^\eta$ ,  $B_{l'm}^\phi$  surface integrals of Eqs. (34)–(37) are determined by numerical integration. The linear algebraic equations of Eqs. (26)–(33) are then solved for the internal and scattered field coefficients ( $a_{lm}$ ,  $b_{lm}$ ,  $c_{lm}$ ,  $d_{lm}$ ,  $e_{lm}$ ,  $f_{lm}$ ,  $g_{lm}$ ,  $h_{lm}$ ). Once the expansion coefficients are known, Eqs. (8)–(17) can then be used to determine the electromagnetic field anywhere inside or outside the spheroid.

### B. Prolate Geometry

In prolate spheroidal coordinates, the rectangular coordinates ( $x$ ,  $y$ ,  $z$ ) are related to the spheroidal coordinates ( $-1 < \eta < +1$ ,  $1 < \xi < \infty$ ,  $0 < \phi < 2\pi$ ) by

$$x = [(\xi^2 - 1)(1 - \eta^2)]^{1/2} \cos \phi, \quad (38)$$

$$y = [(\xi^2 - 1)(1 - \eta^2)]^{1/2} \sin \phi, \quad (39)$$

$$z = \xi\eta. \quad (40)$$

The semimajor to semiminor axis ratio of the prolate spheroid ( $a/b$ ) is related to the choice of  $\xi_0$  by

$$(a/b) = \frac{\xi_0}{(\xi_0^2 - 1)^{1/2}}. \quad (41)$$

So, for example,  $\xi_0 = 1.15470 \Rightarrow (a/b) = 2$ ,  $\xi_0 = 1.06066 \Rightarrow (a/b) = 3$ , etc.

The components of the eigenfunctions of the vector Helmholtz equation for a prolate spheroidal coordinate system are given in Ref. 1.

### C. Oblate Geometry

In oblate spheroidal coordinates, the rectangular coordinates ( $x$ ,  $y$ ,  $z$ ) are related to the spheroidal coordinates ( $-1 < \eta < +1$ ,  $0 < \xi < \infty$ ,  $0 < \phi < 2\pi$ ) by

$$x = [(\xi^2 + 1)(1 - \eta^2)]^{1/2} \cos \phi, \quad (42)$$

$$y = [(\xi^2 + 1)(1 - \eta^2)]^{1/2} \sin \phi, \quad (43)$$

$$z = \xi\eta. \quad (44)$$

The semimajor to semiminor axis ratio of the oblate spheroid ( $a/b$ ) is related to the choice of  $\xi_0$  by

$$(a/b) = (1 + 1/\xi_0^2)^{1/2}. \quad (45)$$

So, for example,  $\xi_0 = 0.57735 \Rightarrow (a/b) = 2$ ,  $\xi_0 = 0.35355 \Rightarrow (a/b) = 3$ , etc.

The components of the eigenfunctions of the vector Helmholtz equation for an oblate spheroidal coordinate system are given in Ref. 1.

### 3. Verification Calculations

Computer programs were written based on the theoretical development discussed in Section 2. I verified these computer programs (and the inherent theoretical development) by performing several tests. Electromagnetic field distributions were generated for layered prolate and oblate spheroids of a 1.3 axis ratio, and the results were compared point by point with corresponding results obtained by use of the spherical coordinate boundary-matching method of Ref. 11. For the limited axis ratio of 1.3, the boundary-matching method is rigorously valid, and the results of the spheroidal separation-of-variables solution and the spherical coordinate boundary-matching method solution were found to be identical to the order of six significant digits. It was also verified that the homogeneous spheroid solution (as given in Ref. 1) could be recovered from the layered spheroid solution when either the relative index of refraction of the layer was set equal to the relative index of refraction of the core ( $n_l = n_c$ ) or when the relative index of refraction of the layer was set equal to unity ( $n_l = 1$ ).

Finally, I obtained an additional verification by checking the rigor with which the electromagnetic boundary conditions are maintained over the outer surface of the spheroid and over the surface of the core-layer interface. These boundary conditions include not only the imposed tangential conditions of Eqs. (18)–(25), but also the following supplementary

conditions on the normal components of the electromagnetic field. At  $\xi = \xi_0$ ,

$$E_{\xi}^{(i)} + E_{\xi}^{(s)} = n_l^2 E_{\xi}^{(l)}, \quad (46)$$

$$H_{\xi}^{(i)} + H_{\xi}^{(s)} = H_{\xi}^{(l)}; \quad (47)$$

and at  $\xi = \xi_c$ ,

$$n_l^2 E_{\xi}^{(l)} = n_c^2 E_{\xi}^{(c)}, \quad (48)$$

$$H_{\xi}^{(l)} = H_{\xi}^{(c)}. \quad (49)$$

The absolute values of the deviations of the left- and right-hand sides of Eqs. (46) and (47), averaged over the outer surface of the spheroid, and the absolute values of the deviations of the left- and right-hand sides of Eqs. (48) and (49), averaged over the surface of the core-layer interface, provided sensitive criteria for convergence. Normal convergence was observed, and the two convergence criteria could be made negligibly small when the summation index limits  $M$  and  $L$  were chosen to be sufficiently large.

#### 4. Demonstration Calculations

A set of calculations demonstrating the effects of the presence of the layer on the internal, near-surface, and scattered electromagnetic fields of the spheroid was performed. For the calculations that follow, the external dielectric constant ( $\epsilon_{\text{ext}}$ ) was set equal to 1.0, and the values of 1.33 and 1.50 were used for the relative index of refractions of the core ( $n_c$ ) and the layer ( $n_l$ ). A 2:1 axis ratio spheroid ( $\xi_0 = 1.154701$  for prolate,  $\xi_0 = 0.577350$  for oblate) with a 2.5:1 axis ratio core ( $\xi_c = 1.091089$  for prolate,  $\xi_0 = 0.436436$  for oblate) was assumed. For these conditions, 54.0% of the total volume of the prolate spheroid is in the layer, whereas for the oblate spheroid 67.5% of the total volume is in the layer. The spheroid size parameter ( $h_{\text{ext}}$ ) was set to a value ( $h_{\text{ext}} = 27.494593$  for prolate,  $h_{\text{ext}} = 21.822473$  for oblate) corresponding to an equivalent sphere (i.e., equal volume sphere) size parameter of 20.

The incident electromagnetic field, either plane wave or focused Gaussian beam, was assumed to be linearly polarized. The direction of propagation and the direction of electric field polarization are specified with two beam direction angles,  $\theta_{bd}$  and  $\phi_{bd}$ .  $\theta_{bd}$  is the direction of propagation relative to the  $y$ - $z$  plane (see Fig. 1), and  $\phi_{bd}$  is the direction of the incident electric field polarization relative to the  $x$ - $z$  plane.

Figure 2 shows a surface grid plot of the electric field magnitude ( $|\mathbf{E}|$ ) in the  $x$ - $z$  plane for a plane wave incident upon a 2:1 axis ratio layered prolate spheroid with a core relative index of refraction  $n_c$  of 1.33 and a layer relative index of refraction  $n_l$  of 1.50. The angle of incident propagation  $\theta_{bd}$  is  $30^\circ$ , and the angle of incident electric field polarization  $\phi_{bd}$  is  $90^\circ$  (perpendicular to the  $x$ - $z$  plane). (Note: In the plane perpendicular to the direction of the incident electric field polarization, the  $\xi$  component of the electric field is zero, and thus the electric field is continuous across both the outer surface of the spheroid and

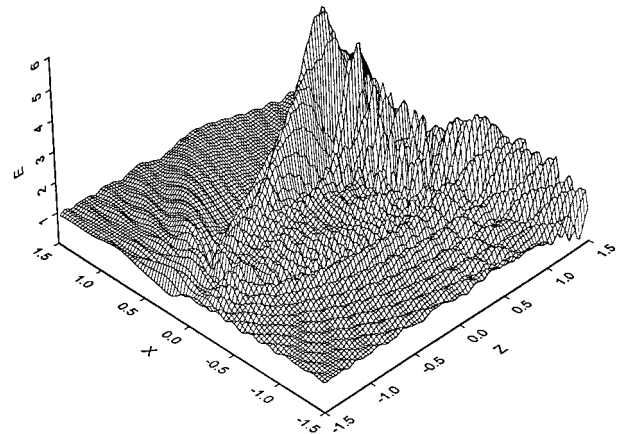


Fig. 2. Surface grid plot of the electric field magnitude in the  $x$ - $z$  plane for a plane wave incident upon a 2:1 axis ratio prolate spheroid with a 2.5:1 axis ratio core. Incident propagation angle  $\theta_{bd} = 30^\circ$ , incident electric field polarization angle  $\phi_{bd} = 90^\circ$ , spheroid size parameter  $h_{\text{ext}} = 27.494593$ , relative index of refraction of core  $n_c = 1.33$ , and relative index of refraction of layer  $n_l = 1.50$ .

the surface of the core-layer interface, as shown in Fig. 2.) The plot extends from  $-1.5$  to  $+1.5$  in both the  $x$  and  $z$  directions and includes the electric field of the core, layer, and near surface.

As shown in Fig. 2, part of the incident plane wave apparently reflects and diffracts off the front surface of the prolate spheroid creating lines of constructive and destructive interference in the near field. The remaining part of the incident plane wave is transmitted into the spheroid where it is refracted at the layer-external interface and then subsequently interacts with the core-layer interface. There are internal regions of constructive and destructive interference with several peaks on the shadow side of the spheroid (perhaps created by a focusing of the electromagnetic wave as it passes through the spheroid). Part of the electromagnetic energy is apparently transmitted out the shadow-side end of the spheroid in the form of a relatively well-defined beam. Figure 3 gives contour and gray-level (white implies low, dark implies high) plots of the same data shown in Fig. 2. Both plots in Fig. 3 include lines

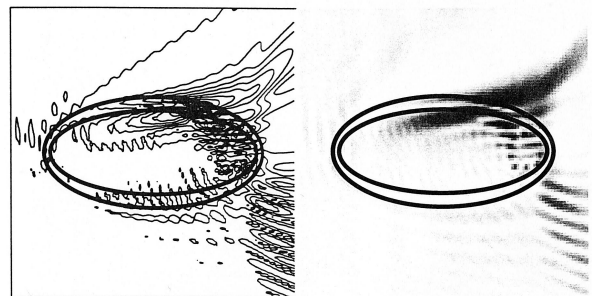


Fig. 3. Contour and gray-level (white implies low, black implies high) plots of the electric field magnitude in the  $x$ - $z$  plane for a plane wave incident upon a 2:1 axis ratio layered prolate spheroid with a 2.5:1 axis ratio core. Same conditions as Fig. 2.

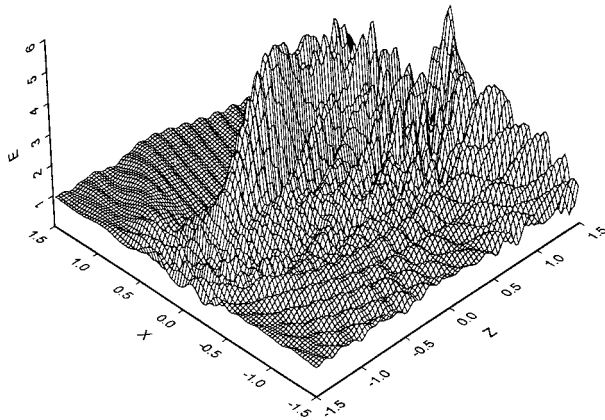


Fig. 4. Surface grid plot of the electric field magnitude in the  $x$ - $z$  plane for a plane wave incident upon a 2:1 axis ratio prolate spheroid with a 2.5:1 axis ratio core.  $\theta_{bd} = 30^\circ$ ,  $\phi_{bd} = 90^\circ$ ,  $h_{ext} = 27.494593$ ,  $n_c = 1.50$ ,  $n_l = 1.33$ .

showing the locations of the spheroid outer surface and core-layer interface boundaries.

Figure 4 provides a surface grid plot of the electric field magnitude for the same layered prolate spheroid conditions as in Fig. 2, except that the relative index of refractions of the layer and core were interchanged so that now  $n_c = 1.50$  and  $n_l = 1.33$ . As can be seen by a comparison with Figs. 2 and 4, the internal electric field distribution is significantly affected by the interchange of the two index of refraction values. Figure 5 is a surface grid plot of a corresponding homogeneous spheroid with a uniform relative index of refraction of 1.33 ( $n_c = n_l = 1.33$ ). The electric field distribution for the homogeneous spheroid with  $n = 1.33$  is similar to (but not identical to) the electric field distribution of the  $n_c = 1.33$ ,  $n_l = 1.50$  layered spheroid given in Fig. 2. (The electric field distribution of a homogeneous spheroid with  $n = 1.50$ , not shown, was found to be similar to that of the  $n_c = 1.50$ ,  $n_l = 1.33$  layered spheroid given in Fig. 4.)

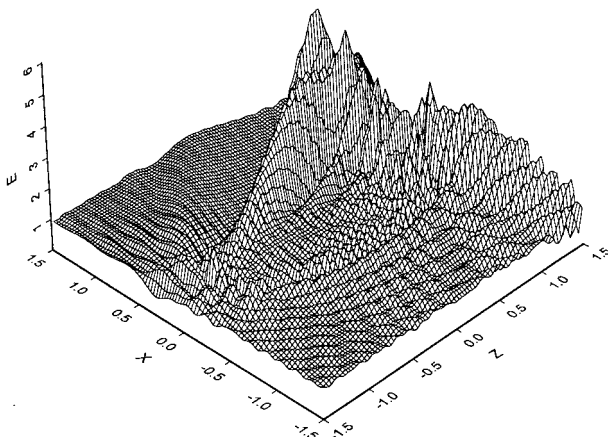


Fig. 5. Surface grid plot of the electric field magnitude in the  $x$ - $z$  plane for a plane wave incident upon a 2:1 axis ratio homogeneous prolate spheroid.  $\theta_{bd} = 30^\circ$ ,  $\phi_{bd} = 90^\circ$ ,  $h_{ext} = 27.494593$ ,  $n_c = n_l = 1.33$ .

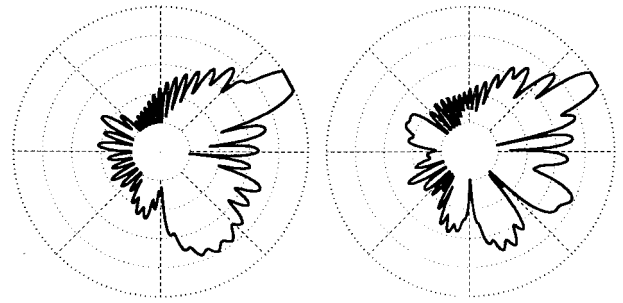


Fig. 6. Polar plots (log scale, range from 0.001 to 10.0) of the far-field scattering in the  $x$ - $z$  plane for a plane wave incident upon a 2:1 axis ratio prolate spheroid with a 2.5:1 axis ratio core.  $\theta_{bd} = 30^\circ$ ,  $\phi_{bd} = 90^\circ$ ,  $h_{ext} = 27.494593$ . For the plot on the left,  $n_c = 1.33$  and  $n_l = 1.50$  ( $S_{r,max} = 18.77495$ ); for the plot on the right,  $n_c = 1.50$  and  $n_l = 1.33$  ( $S_{r,max} = 14.79815$ ).

Apparently, at least for the conditions considered here, even though more than 50% of the spheroid volume is in the layer, the internal and near-surface electric field distribution is most strongly affected by the optical properties of the core, as opposed to the optical properties of the layer.

We can determine the far-field scattering intensity by evaluating the scattered field in the limit of  $\xi \Rightarrow \infty$ . For the spheroidal coordinate system—in the limit of a large spheroidal radial coordinate ( $\xi \Rightarrow \infty$ )—the constant  $\xi$  surface becomes spherical, and spheroidal coordinates can be directly related to the corresponding spherical coordinates:  $\xi f \Rightarrow r$  and  $\eta \Rightarrow \cos(\theta)$ . A nondimensionalized, time-averaged scattering intensity in the far field can then be defined as follows:

$$S_r(\theta, \phi) = \lim_{r \rightarrow \infty} \frac{r^2 \langle \mathbf{S} \rangle_r}{\frac{c}{8\pi} E_0^2 (\pi f^2)} = \lim_{\xi \rightarrow \infty} \frac{\xi^2}{\pi} \text{Re}[E_\phi^{(s)} H_\eta^{(s)*} - E_\eta^{(s)} H_\phi^{(s)*}]. \quad (50)$$

Figure 6 shows a polar plot (log scale) of the far-field scattering in the  $x$ - $z$  plane for the layered prolate spheroid. The plot on the left is for  $n_c = 1.33$  and  $n_l = 1.50$  (corresponding to Fig. 2), and the plot on the right is for  $n_c = 1.50$  and  $n_l = 1.33$  (corresponding to Fig. 4). Both scattering patterns exhibit a strong forward scattering lobe at  $30^\circ$ , which corresponds with the propagation direction of the incident plane wave. The scattering pattern is nonsymmetrical about the direction of incident propagation, with relatively strong scattering in the  $-60^\circ$  region and relatively weak scattering in the  $120^\circ$  region. The far-field scattering patterns for homogeneous prolate spheroids with  $n = 1.33$  and  $n = 1.50$  are shown in Fig. 7, and the scattering patterns of the homogeneous spheroids exhibit similar general features to those of the layered prolate spheroids. Although the scattering patterns of the four types of prolate spheroid have similar general features, there are differences, and an analysis of these differences

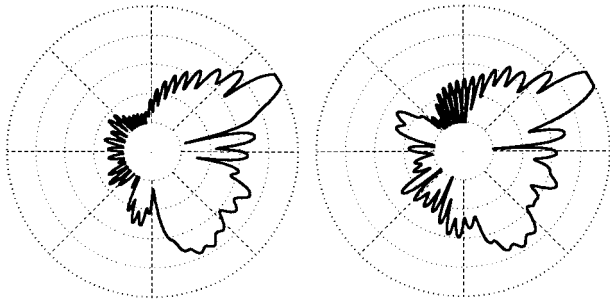


Fig. 7. Polar plots (log scale, range from 0.001 to 10.0) of the far-field scattering in the  $x$ - $z$  plane for a plane wave incident upon a 2:1 axis ratio homogeneous prolate spheroid.  $\theta_{bd} = 30^\circ$ ,  $\phi_{bd} = 90^\circ$ ,  $h_{ext} = 27.494593$ . For the plot on the left,  $n_c = n_l = 1.33$  ( $S_{r,max} = 12.18744$ ); for the plot on the right,  $n_c = n_l = 1.50$  ( $S_{r,max} = 14.48922$ ).

might be necessary, for example, to assist the development of a light-scattering measurement instrument capable of distinguishing one type of prolate spheroid from another.

All results so far have been for the prolate geometry. Figure 8 shows a surface grid plot of the electric field magnitude ( $|\mathbf{E}|$ ) in the  $x$ - $z$  plane for a plane wave incident upon a 2:1 axis ratio layered oblate spheroid with a core relative index of refraction  $n_c$  of 1.33 and a layer relative index of refraction  $n_l$  of 1.50. As was the case for the prolate calculations, the angle of incident propagation  $\theta_{bd}$  is  $30^\circ$ , and the angle of incident electric field polarization  $\phi_{bd}$  is  $90^\circ$  (perpendicular to the  $x$ - $z$  plane). The electric field distribution for the oblate spheroid is quite different from that of the prolate spheroid. The most salient feature is the strong focusing peak that occurs on the shadow side of the spheroid. Contour and gray-level plots of the same data presented in Fig. 8 are given in Fig. 9. Figure 10 provides the electric field distribution for the same layered oblate spheroid, but with the values

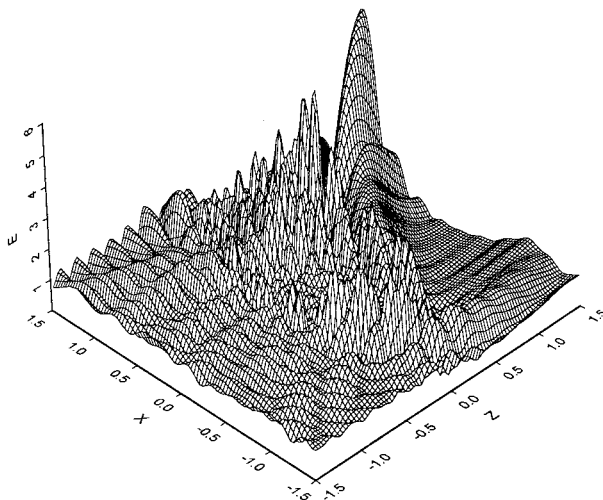


Fig. 8. Surface grid plot of the electric field magnitude in the  $x$ - $z$  plane for a plane wave incident upon a 2:1 axis ratio oblate spheroid with a 2.5:1 axis ratio core.  $\theta_{bd} = 30^\circ$ ,  $\phi_{bd} = 90^\circ$ ,  $h_{ext} = 21.822473$ ,  $n_c = 1.33$ ,  $n_l = 1.50$ .

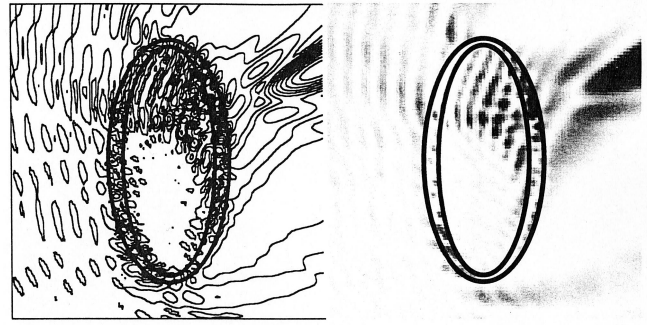


Fig. 9. Contour and gray-level (white implies low, black implies high) plots of the electric field magnitude in the  $x$ - $z$  plane for a plane wave incident upon a 2:1 axis ratio layered oblate spheroid with a 2.5:1 axis ratio core. Same conditions as Fig. 8.

of the core and layer relative index of refractions interchanged ( $n_c = 1.50$ ,  $n_l = 1.33$ ); and Fig. 11 gives the electric field distribution for the corresponding homogeneous oblate spheroid ( $n_c = n_l = 1.33$ ). In a comparison of Figs. 8, 10, and 11, it can be observed that, although the general features of the electric field distributions are similar, they are not identical. The corresponding far-field scattering patterns of the layered and homogeneous oblate spheroids are shown, respectively, in Figs. 12 and 13. The scattering patterns are quite distinct from those of the prolate spheroid, and, although once again exhibiting similar general features, each scattering pattern has distinguishing features that could be used to identify one type of oblate spheroid from the other.

Solutions can be obtained for any known incident field, not just the plane wave. As an example, Fig. 14 shows a surface grid plot of the electric field distribution for the same layered oblate spheroid considered in Fig. 8, but now with focused beam illumination instead of plane-wave illumination. The corresponding contour and gray-level plots are

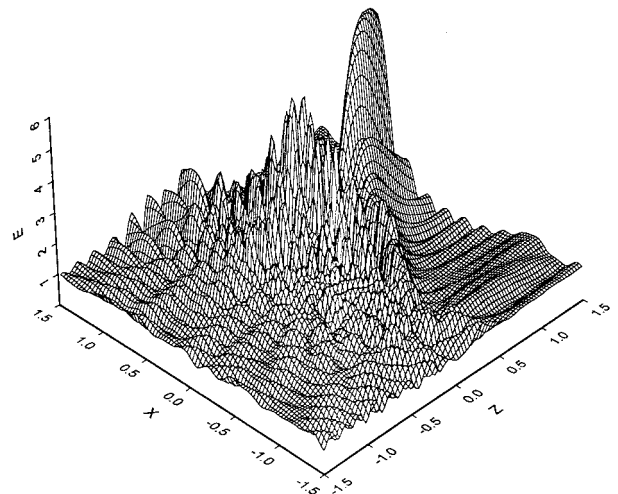


Fig. 10. Surface grid plot of the electric field magnitude in the  $x$ - $z$  plane for a plane wave incident upon a 2:1 axis ratio oblate spheroid with a 2.5:1 axis ratio core.  $\theta_{bd} = 30^\circ$ ,  $\phi_{bd} = 90^\circ$ ,  $h_{ext} = 21.822473$ ,  $n_c = 1.50$ ,  $n_l = 1.33$ .

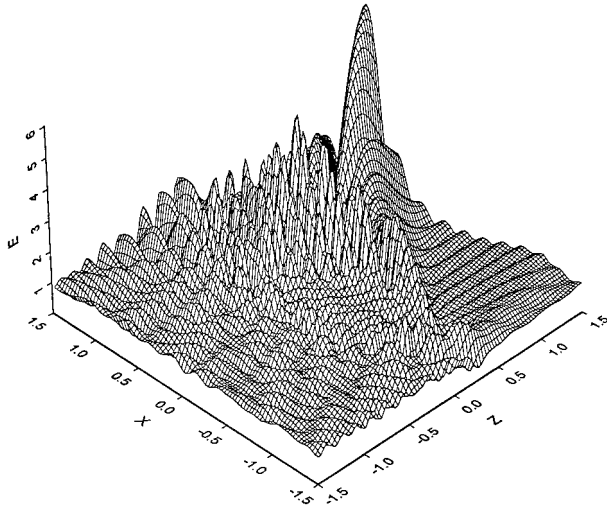


Fig. 11. Surface grid plot of the electric field magnitude in the  $x$ - $z$  plane for a plane wave incident upon a 2:1 axis ratio homogeneous oblate spheroid.  $\theta_{bd} = 30^\circ$ ,  $\phi_{bd} = 90^\circ$ ,  $h_{ext} = 21.822473$ ,  $n_c = n_l = 1.33$ .

given in Fig. 15. The beam propagates in the  $z$ -axis direction ( $\theta_{bd} = 0^\circ$ ) with electric field polarization

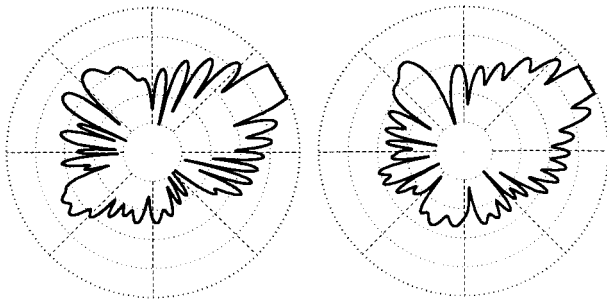


Fig. 12. Polar plots (log scale, range from 0.001 to 10.0) of the far-field scattering in the  $x$ - $z$  plane for a plane wave incident upon a 2:1 axis ratio oblate spheroid with a 2.5:1 axis ratio core.  $\theta_{bd} = 30^\circ$ ,  $\phi_{bd} = 90^\circ$ ,  $h_{ext} = 21.822473$ . For the plot on the left,  $n_c = 1.33$  and  $n_l = 1.50$  ( $S_{r,max} = 84.95962$ ); for the plot on the right,  $n_c = 1.50$  and  $n_l = 1.33$  ( $S_{r,max} = 64.08979$ ).

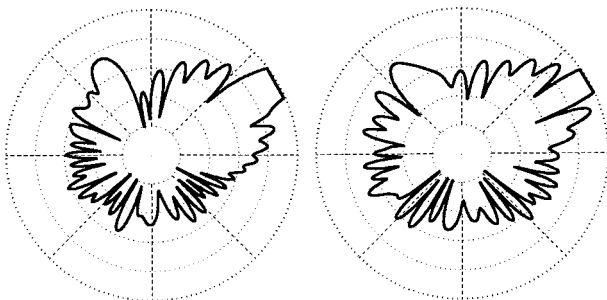


Fig. 13. Polar plots (log scale, range from 0.001 to 10.0) of the far-field scattering in the  $x$ - $z$  plane for a plane wave incident upon a 2:1 axis ratio homogeneous oblate spheroid.  $\theta_{bd} = 30^\circ$ ,  $\phi_{bd} = 90^\circ$ ,  $h_{ext} = 21.822473$ . For the plot on the left,  $n_c = n_l = 1.33$  ( $S_{r,max} = 60.93952$ ); for the plot on the right,  $n_c = n_l = 1.50$  ( $S_{r,max} = 51.01663$ ).

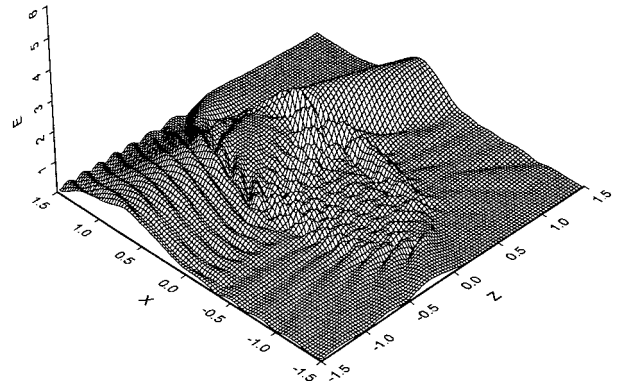


Fig. 14. Surface grid plot of the electric field magnitude in the  $x$ - $z$  plane for a focused beam incident upon a 2:1 axis ratio oblate spheroid with a 2.5:1 axis ratio core.  $\theta_{bd} = 30^\circ$ ,  $\phi_{bd} = 90^\circ$ ,  $h_{ext} = 21.822473$ ,  $n_c = 1.33$ ,  $n_l = 1.50$ ,  $w_0 = 0.50$ ,  $(x_0, y_0, z_0) = (0.0, 0.0, 0.75)$ .

perpendicular to the  $x$ - $z$  plane ( $\phi_{bd} = 90^\circ$ ). The beam waist is equal to one half of the semifocal length of the spheroid ( $w_0 = 0.5$ ), and the focal point is located on the  $x$  axis at  $z_0 = 0.75$ . The electromagnetic field components of the incident beam were evaluated by use of the fifth-order corrected fundamental Gaussian (i.e., TEM<sub>00</sub> mode) beam model described in Ref. 15. As can be seen in Figs. 14 and 15, the layered oblate spheroid acts as a lens, with the beam entering the illuminated side and then exiting the shadow side at an angle of approximately 30 deg relative to the original direction of propagation.

## 5. Summary

A spheroidal coordinate separation-of-variables solution has been developed, verified, and demonstrated for determination of the internal, near-surface, and scattered fields of a layered spheroid (either prolate or oblate) with arbitrary monochromatic illumination. The demonstration calculations illustrate how the presence of the layer can affect the internal and near fields, as well as affect the resultant far-field scattering pattern. A possible future application of this research would be to perform additional systematic calculations to develop a better understanding of the combined effects of the layered spheroid geome-

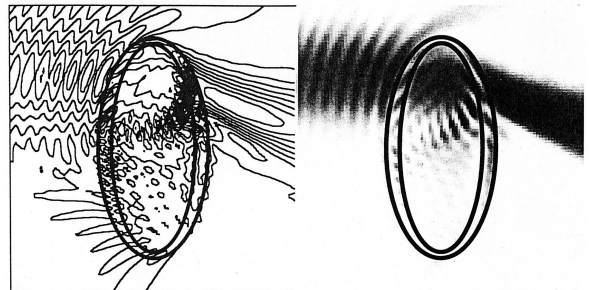


Fig. 15. Contour and gray-level (white implies low, black implies high) plots of the electric field magnitude in the  $x$ - $z$  plane for a focused beam incident upon a 2:1 axis ratio layered oblate spheroid with a 2.5:1 axis ratio core. Same conditions as Fig. 14.

try, layered spheroid optical properties, and incident field properties on far-field scattering.

#### Appendix A: One-Dimensional Integrals

$$I_{lml'}^1 = 2 \int_0^1 N_{lm,\eta}^{(s)}(\xi_0, \eta, 0) S_{l',m}(h_{\text{ext}}, \eta) d\eta, \quad (\text{A1})$$

$$I_{lml'}^2 = 2 \int_0^1 M_{lm,\eta}^{(s)}(\xi_0, \eta, 0) S_{l',m}(h_{\text{ext}}, \eta) d\eta, \quad (\text{A2})$$

$$I_{lml'}^3 = 2 \int_0^1 N_{lm,\eta}^{(l,1)}(\xi_0, \eta, 0) S_{l',m}(h_{\text{ext}}, \eta) d\eta, \quad (\text{A3})$$

$$I_{lml'}^4 = 2 \int_0^1 M_{lm,\eta}^{(l,1)}(\xi_0, \eta, 0) S_{l',m}(h_{\text{ext}}, \eta) d\eta, \quad (\text{A4})$$

$$I_{lml'}^5 = 2 \int_0^1 N_{lm,\eta}^{(l,2)}(\xi_0, \eta, 0) S_{l',m}(h_{\text{ext}}, \eta) d\eta, \quad (\text{A5})$$

$$I_{lml'}^6 = 2 \int_0^1 M_{lm,\eta}^{(l,2)}(\xi_0, \eta, 0) S_{l',m}(h_{\text{ext}}, \eta) d\eta, \quad (\text{A6})$$

$$I_{lml'}^7 = 2 \int_0^1 N_{lm,\phi}^{(s)}(\xi_0, \eta, 0) S_{l',m}(h_{\text{ext}}, \eta) d\eta, \quad (\text{A7})$$

$$I_{lml'}^8 = 2 \int_0^1 M_{lm,\phi}^{(s)}(\xi_0, \eta, 0) S_{l',m}(h_{\text{ext}}, \eta) d\eta, \quad (\text{A8})$$

$$I_{lml'}^9 = 2 \int_0^1 N_{lm,\phi}^{(l,1)}(\xi_0, \eta, 0) S_{l',m}(h_{\text{ext}}, \eta) d\eta, \quad (\text{A9})$$

$$I_{lml'}^{10} = 2 \int_0^1 M_{lm,\phi}^{(l,1)}(\xi_0, \eta, 0) S_{l',m}(h_{\text{ext}}, \eta) d\eta, \quad (\text{A10})$$

$$I_{lml'}^{11} = 2 \int_0^1 N_{lm,\phi}^{(l,2)}(\xi_0, \eta, 0) S_{l',m}(h_{\text{ext}}, \eta) d\eta, \quad (\text{A11})$$

$$I_{lml'}^{12} = 2 \int_0^1 M_{lm,\phi}^{(l,2)}(\xi_0, \eta, 0) S_{l',m}(h_{\text{ext}}, \eta) d\eta, \quad (\text{A12})$$

$$I_{lml'}^{13} = 2 \int_0^1 N_{lm,\eta}^{(c)}(\xi_c, \eta, 0) S_{l',m}(h_l, \eta) d\eta, \quad (\text{A13})$$

$$I_{lml'}^{14} = 2 \int_0^1 M_{lm,\eta}^{(c)}(\xi_c, \eta, 0) S_{l',m}(h_l, \eta) d\eta, \quad (\text{A14})$$

$$I_{lml'}^{15} = 2 \int_0^1 N_{lm,\eta}^{(l,1)}(\xi_c, \eta, 0) S_{l',m}(h_l, \eta) d\eta, \quad (\text{A15})$$

$$I_{lml'}^{16} = 2 \int_0^1 M_{lm,\eta}^{(l,1)}(\xi_c, \eta, 0) S_{l',m}(h_l, \eta) d\eta, \quad (\text{A16})$$

$$I_{lml'}^{17} = 2 \int_0^1 N_{lm,\eta}^{(l,2)}(\xi_c, \eta, 0) S_{l',m}(h_l, \eta) d\eta, \quad (\text{A17})$$

$$I_{lml'}^{18} = 2 \int_0^1 M_{lm,\eta}^{(l,2)}(\xi_c, \eta, 0) S_{l',m}(h_l, \eta) d\eta, \quad (\text{A18})$$

$$I_{lml'}^{19} = 2 \int_0^1 N_{lm,\phi}^{(c)}(\xi_c, \eta, 0) S_{l',m}(h_l, \eta) d\eta, \quad (\text{A19})$$

$$I_{lml'}^{20} = 2 \int_0^1 M_{lm,\phi}^{(c)}(\xi_c, \eta, 0) S_{l',m}(h_l, \eta) d\eta, \quad (\text{A20})$$

$$I_{lml'}^{21} = 2 \int_0^1 N_{lm,\phi}^{(l,1)}(\xi_c, \eta, 0) S_{l',m}(h_l, \eta) d\eta, \quad (\text{A21})$$

$$I_{lml'}^{22} = 2 \int_0^1 M_{lm,\phi}^{(l,1)}(\xi_c, \eta, 0) S_{l',m}(h_l, \eta) d\eta, \quad (\text{A22})$$

$$I_{lml'}^{23} = 2 \int_0^1 N_{lm,\phi}^{(l,2)}(\xi_c, \eta, 0) S_{l',m}(h_l, \eta) d\eta, \quad (\text{A23})$$

$$I_{lml'}^{24} = 2 \int_0^1 M_{lm,\phi}^{(l,2)}(\xi_c, \eta, 0) S_{l',m}(h_l, \eta) d\eta, \quad (\text{A24})$$

This research was supported, in part, by the Edge-wood Chemical Biological Center (Merrill E. Milham) under the auspices of the U.S. Army Research Office Scientific Services Program administrated by Battelle (delivery order 258, contract DAAH04-96-C-0086).

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